

PYTHON FUNCTIONS — COMPLETE NOTES

Beginner-Friendly Theory, Examples, Practice Questions and Solutions

1. What is a Function?

A function is a reusable block of code that performs a specific task. Instead of writing the same code again and again, we define it once and call it whenever needed.

Example:

```
def greet():  
    print("Hello")  
  
greet()
```

Output:

```
Hello
```

2. Why Do We Use Functions?

- Code reusability: write code once and use it many times.
- Reduces repetition: avoids writing the same statements repeatedly.
- Improves readability: makes a program easier to understand.
- Makes debugging easier: problems can be isolated inside individual functions.
- Makes large programs easier to organize and maintain.

3. Function Definition and Function Call

A function has two important parts:

1. Function definition: creating the function using the def keyword.
2. Function call: executing the function by writing its name followed by parentheses.

Syntax:

```
def function_name():  
    # statements  
  
function_name()
```

Example:

```
def welcome():  
    print("Welcome to Python")  
  
welcome()
```

4. Parameters and Arguments

A parameter is a variable written inside the function definition. An argument is the actual value passed to the function when it is called.

```
def greet(name):          # name is a parameter
    print("Hello", name)

greet("Rahul")            # "Rahul" is an argument
```

Output:

```
Hello Rahul
```

5. Function with Multiple Parameters

```
def add(a, b):
    print(a + b)

add(10, 20)
```

Output:

```
30
```

6. return Statement

The return statement sends a value back from a function. The returned value can be stored in a variable or used in another expression.

```
def add(a, b):
    return a + b

result = add(10, 20)
print(result)
```

Output:

```
30
```

7. print() vs return

print()	return	Main Difference
Displays a value on the screen.	Sends a value back from the function.	print is for displaying; return is for giving a result back.
Does not normally give the calling code a usable result.	Can be stored in a variable.	return is useful when the result is needed later.

Example:

```
def square(n):
    return n * n
```

```
answer = square(5)
print(answer)
```

Output:

```
25
```

8. Function Without Parameters

```
def message():
    print("Learning Python")

message()
```

Output:

```
Learning Python
```

9. Function With Parameters

```
def multiply(a, b):
    print(a * b)

multiply(4, 5)
```

Output:

```
20
```

10. Function With a Return Value

```
def multiply(a, b):
    return a * b

result = multiply(4, 5)
print(result)
```

Output:

```
20
```

11. Default Parameters

A default parameter has a value that is automatically used if the caller does not provide an argument.

```
def greet(name="Student"):
    print("Hello", name)

greet()
greet("Aman")
```

Output:

```
Hello Student
Hello Aman
```

12. Keyword Arguments

Keyword arguments are passed using the parameter name. They make function calls clearer and can be written in a different order.

```
def student(name, age):
    print("Name:", name)
    print("Age:", age)

student(age=20, name="Aman")
```

Output:

```
Name: Aman
Age: 20
```

13. Local Variables

A local variable is created inside a function and can normally be used only inside that function.

```
def show():
    x = 10
    print(x)

show()
```

Here, x is a local variable.

14. Global Variables

A global variable is created outside a function and can generally be accessed from different parts of the program.

```
x = 10

def show():
    print(x)

show()
```

Output:

```
10
```

15. Built-in Functions

Python provides many functions that are already available for use.

Function	Purpose
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print()	Displays output
input()	Takes input from the user
len()	Returns the length of an object
type()	Returns the data type
int()	Converts a value to integer
str()	Converts a value to string
sum()	Returns the sum of values
max()	Returns the largest value
min()	Returns the smallest value

16. User-Defined Functions

A user-defined function is a function created by the programmer using the def keyword.

```
def add(a, b):
    return a + b

print(add(5, 7))
```

Output:

12

17. Function with if-else

```
def check_even(num):
    if num % 2 == 0:
        return "Even"
    else:
        return "Odd"

print(check_even(10))
```

Output:

Even

18. Function with Loops

```
def factorial(n):
    fact = 1
    for i in range(1, n + 1):
        fact = fact * i
    return fact

print(factorial(5))
```

Output:

120

19. Real-Life Example: ATM

An ATM can be understood like a function. You select an operation, provide required information, the ATM performs the task, and you receive a result.

Python Concept	ATM Example
Function	ATM withdrawal operation
Argument	Amount such as ₹5000
Function body	ATM checks balance and processes the withdrawal
Return/output	Cash received or a message such as insufficient balance

20. Important Function Terminology

Term	Meaning
Function	Reusable block of code that performs a task
def	Keyword used to define a function
Function call	Statement used to execute a function
Parameter	Variable in the function definition
Argument	Actual value passed during a function call
return	Sends a result back from a function
Local variable	Variable available within its function
Global variable	Variable defined outside functions

21. General Structure of a Function

```
def function_name(parameters):  
    # function body  
    statements  
    return value
```

Not every function needs parameters or a return statement. Both are optional depending on the task.

22. Theory Questions with Answers

What is a function?

A function is a reusable block of code designed to perform a specific task.

Which keyword is used to define a function?

The def keyword.

What is a function call?

It is the statement used to execute a defined function.

What is a parameter?

A variable listed in a function definition.

What is an argument?

The actual value passed to a function during a function call.

What is return?

It sends a value from the function back to the calling code.

Can a function have multiple parameters?

Yes. A function can have multiple parameters.

Can a function return more than one value?

Yes. Python can return multiple values, commonly as a tuple.

What is a default parameter?

A parameter that has a default value used when no argument is supplied.

What is a built-in function?

A function already provided by Python, such as `print()`, `len()`, and `type()`.

What is a user-defined function?

A function created by the programmer using `def`.

23. Practice Coding Questions with Solutions

Q1. Write a function to print your name.

Solution:

```
def show_name():  
    print("Aman")  
  
show_name()
```

Output:

```
Aman
```

Q2. Write a function to add two numbers.

Solution:

```
def add(a, b):  
    return a + b  
  
print(add(10, 20))
```

Output:

```
30
```

Q3. Write a function to check whether a number is even or odd.

Solution:

```
def check(num):  
    if num % 2 == 0:  
        return "Even"  
    else:  
        return "Odd"  
  
print(check(7))
```

Output:

Odd

Q4. Write a function to find the square of a number.

Solution:

```
def square(num):  
    return num * num  
  
print(square(6))
```

Output:

36

Q5. Write a function to find the largest of two numbers.

Solution:

```
def largest(a, b):  
    if a > b:  
        return a  
    else:  
        return b  
  
print(largest(15, 9))
```

Output:

15

Q6. Write a function to calculate the factorial of a number.

Solution:

```
def factorial(n):  
    fact = 1  
    for i in range(1, n + 1):  
        fact = fact * i  
    return fact  
  
print(factorial(5))
```

Output:

Q7. Write a function to check whether a number is positive, negative, or zero.

Solution:

```
def check_number(n):
    if n > 0:
        return "Positive"
    elif n < 0:
        return "Negative"
    else:
        return "Zero"

print(check_number(-4))
```

Output:

Negative

Q8. Write a function to calculate the area of a rectangle.

Solution:

```
def area(length, width):
    return length * width

print(area(10, 5))
```

Output:

50

Q9. Write a function to print numbers from 1 to 5.

Solution:

```
def print_numbers():
    for i in range(1, 6):
        print(i)

print_numbers()
```

Output:

1\n2\n3\n4\n5

Q10. Write a function to count the number of vowels in a word.

Solution:

```
def count_vowels(word):
    count = 0
    for ch in word.lower():
        if ch in "aeiou":
            count += 1
    return count
```

```
print(count_vowels("python programming"))
```

Output:

4

24. Quick Revision

- Function = reusable block of code.
- Use def to define a function.
- Use the function name with parentheses to call it.
- Parameters are variables in the function definition.
- Arguments are actual values passed to the function.
- return sends a result back to the calling code.
- Functions may have zero, one, or multiple parameters.
- Default parameters provide fallback values.
- Python has built-in functions such as print(), len(), and type().
- User-defined functions are created by the programmer.
- Functions can contain if-else statements and loops.

25. Exam/Viva Tips

- Remember the difference between a parameter and an argument.
- Be able to explain def, function call, and return.
- Know the difference between print() and return.
- Practice writing small functions without looking at the syntax.
- Understand local and global variables at a basic level.
- Practice combining functions with if-else and loops.